Gary D Alvarado

San Francisco Bay Area | galvarado1501@gmail.com | LinkedIn: Gary Alvarado | Github: Denilson15

EDUCATION

University of California, Davis — **Davis, California**
M.S. in Computer Science — *September 2026 - May 2028*

- **Graduate Coursework (in progress):** Computer Vision, Programming Languages

California State University, East Bay — **Hayward, California**
Upper-Division Computer Science Coursework — *May 2025 - May 2026*
B.S. in Business Administration — *Graduated, Fall 2023*

- **Computer Science Coursework:** Software Engineering, Operating Systems, Computer Networks, Analysis of Algorithms, Artificial Intelligence, Computer Architecture, Programming Languages, Data Structures & Algorithms
- **Business Administration Coursework:** IT Management, HR Analytics, Strategic Business Management, Consumer Behavior, Decision Science, Marketing Research, Employee Relations & Law, Operations Management

EXPERIENCE

AIC at UC Davis | *Developer, Advanced Track* — *October 2026 – Present*

Hayward Unified School District | *Substitute Teacher (District-wide)* — *2022 – Present*

- Teach classes across grade levels and subjects district-wide, adapting to new classrooms and lesson plans daily.

Long Term Substitute Teacher | AP Computer Science Principles & Computer Programming — *September 2024 – July 2025*

- Taught 130+ students across 5 sections, covering C++, JavaScript, HTML/CSS, networking, and data representation.
- Designed hands-on projects and lessons that prepared students for the AP CS exam and built problem-solving skills.
- Introduced students to industry tools including GitHub and AI-assisted development.

PROJECTS

Communication System – Java, TCP Sockets, Multithreading, Swing | *May 2026* | Live Demo

- Co-developed a Java client-server chat system in a 5-person team, supporting concurrent users and group chats.
- Built the networking foundation: TCP sockets, a thread-per-client server, and the message protocol and routing.
- Implemented login, account creation, group chats, and message delivery, and connected them to the Swing GUI.
- Co-authored the Software Requirements Specification (SRS) and initial design documentation.

Quizzical – React, JavaScript, API Integration | *January 2026* | Live Demo

- Built an interactive quiz application using React, leveraging state management and hooks for dynamic UI updates.
- Integrated an external trivia API to fetch and display randomized questions and track user responses.
- Implemented conditional rendering and user interaction logic to evaluate answers and display scores.

OCaml Stack-Based Interpreter – OCaml | *August 2025*

- Built a functional interpreter that parses and executes stack-based commands such as Push, Pop, and arithmetic.
- Utilized recursion, pattern matching, and immutable data structures for execution logic.

AP Computer Science Student Portfolio Showcase – JavaScript / HTML / CSS | *May 2025*

- Developed a portfolio platform showcasing 70+ student-built JavaScript projects with links to repos and live sites.

BFS Flight Path Finder – C++ | *April 2025*

- Developed a program that reads a .txt file of city-to-city flights and uses Breadth-First Search (BFS) to determine if a path exists between two locations.
- Demonstrated proficiency in file I/O, graph data structures, and queue-based traversal algorithms.

Interactive Brain Map – JavaScript, Python, AI-Assisted Development | *October 2026* | Live Demo

- Designed and deployed a 3D interactive brain study tool on GitHub Pages using AI-assisted development; users rotate, explode and drill into 79 brain regions through a clickable hierarchy.
- Mapped 81 psychological disorders and neurological diseases to the brain areas they affect, with in-depth write-ups, a 400+ term tap-to-define dictionary, and full English/Spanish support.
- Drove development through repeated user testing, resolving mobile layout, transparency rendering and label-clutter issues across desktop and phone.

SKILLS

Programming: Python, C++, Rust, Java, JavaScript, React, OCaml, SQL, HTML, CSS
Tools: AI, Github, REST APIs, Netlify, Linux, Microsoft Office, Google Classroom